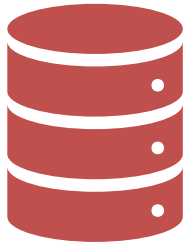


Smart Parking Management System

Cloud-Based Multi-API Integrated Application

Author: Meenakshi (24235415)

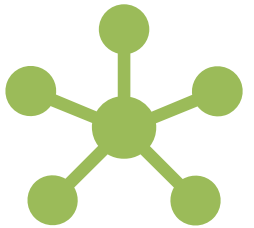
Introduction



Modern applications use microservices over monolithic architecture

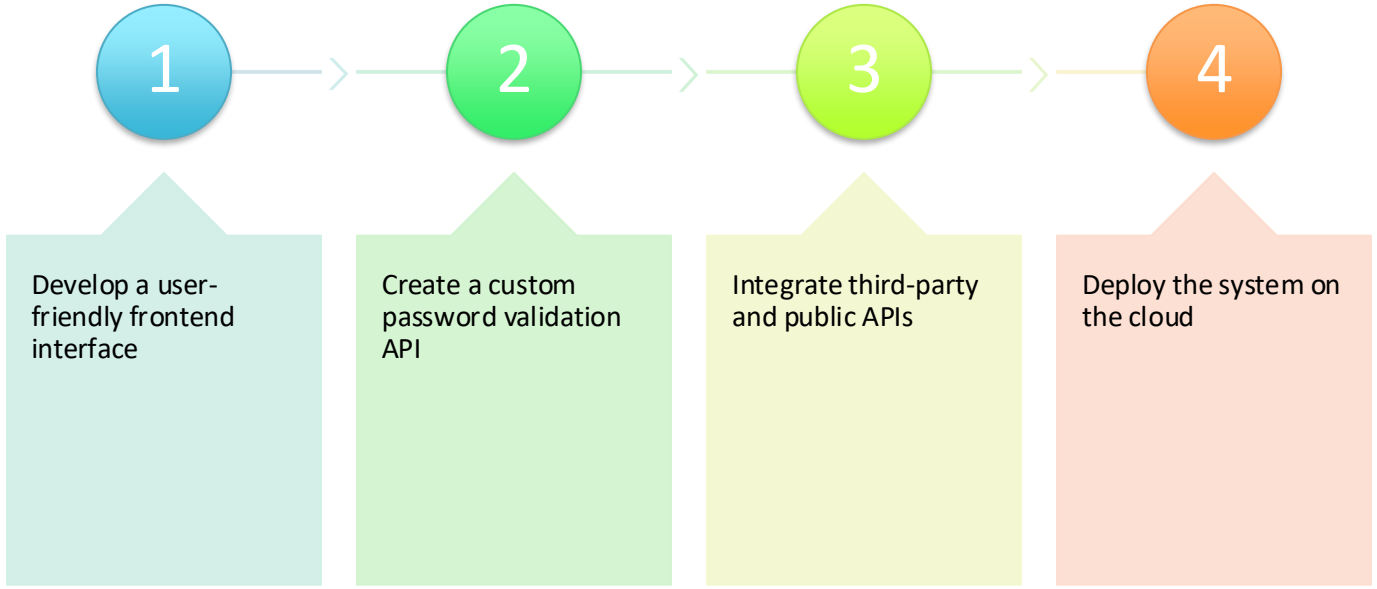


Benefits: Scalability, Flexibility, Maintainability



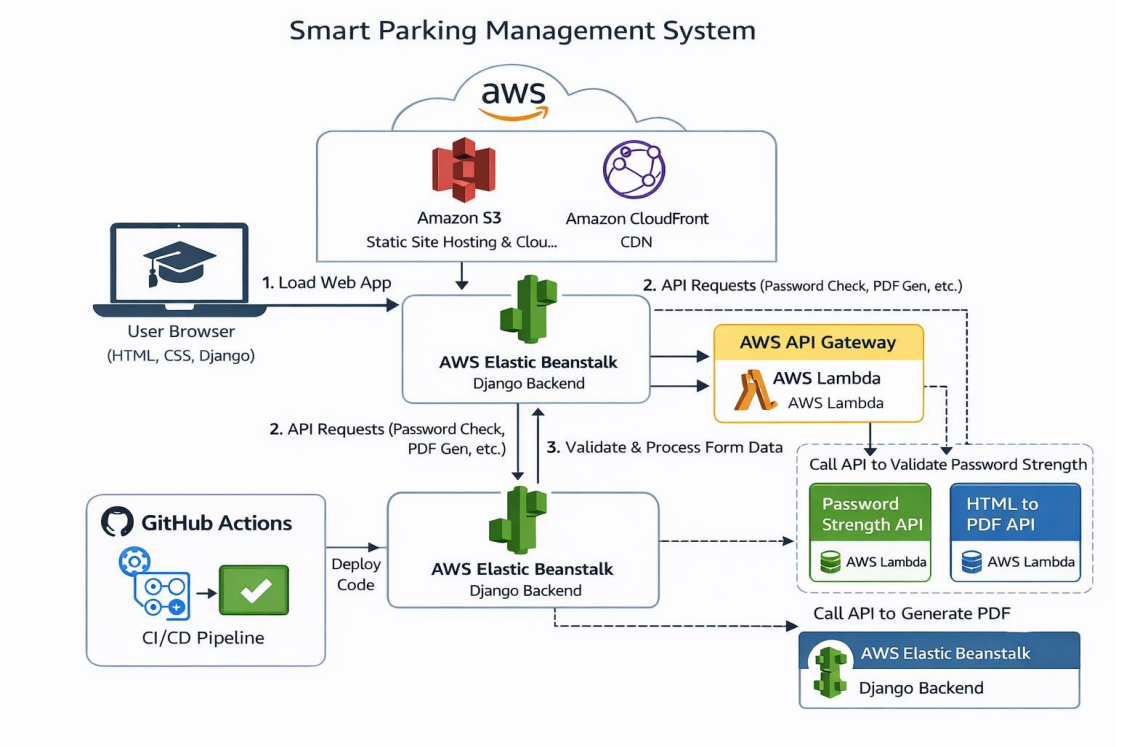
Challenges: API communication, Distributed complexity

Project Objectives



System Overview

- Frontend interacts with users and sends HTTP requests
- Backend processes requests and coordinates with APIs
- APIs handle specific tasks like password validation and PDF generation



Requirements



Functional Requirements:



- Add/Delete vehicle records



- Validate password strength



- Generate PDF of vehicle details



Non-Functional Requirements:



- Scalability, Performance, Availability, Usability

Design Approach



USE OF RESTFUL APIS
FOR COMMUNICATION



STATELESS BACKEND
DESIGN FOR SCALABILITY



LOOSELY COUPLED
SERVICES FOR FLEXIBILITY

Scalability



Cloud deployment enables horizontal scaling



Each service can scale independently



Supports multiple concurrent users

Frontend & Backend



FRONTEND: USER INTERFACE FOR MANAGING VEHICLES



BACKEND: HANDLES LOGIC, API CALLS, AND DATA PROCESSING



COMMUNICATION VIA HTTP REQUESTS

API Integration



CUSTOM API FOR PASSWORD STRENGTH VALIDATION



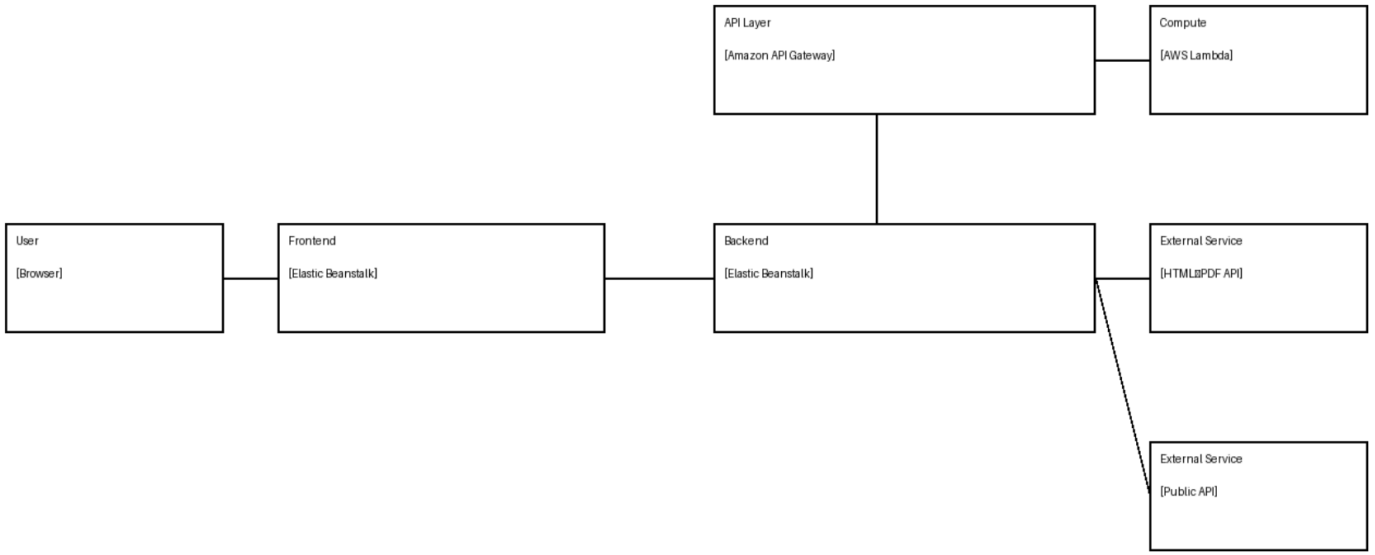
EXTERNAL API FOR HTML-TO-PDF CONVERSION



PUBLIC API INTEGRATED FOR ENHANCED FUNCTIONALITY

Working Flow

- User inputs vehicle data via frontend
- Backend processes request and stores data
- On deletion, backend fetches data and sends to PDF API
- PDF is generated and stored or downloaded



Deployment & CI/CD



Application hosted on cloud platform



Automated deployment using GitHub Actions



Continuous Integration and Delivery pipeline

Conclusion



Learned integration of multiple APIs



Gained experience in cloud deployment



Developed understanding of scalable system design